ISB 140/DS 110−INTRODUCTION TO STATISTICS

LECTURE NOTES

UNIT IV: REGRESSION AND CORRELATION ANALYSIS

1. INTRODUCTION

- ✓ To investigate/study the relationship between two variables we can either use simple linear regression analysis or correlation analysis, one of which is dependent and the other independent.
- ✓ Regression analysis is a statistical technique/tool for investigating and modeling the relationship between variables.
- ✓ Regression analysis is applied in many fields, such as; engineering, the physical and chemical sciences, economics, management, life and biological sciences, and the social sciences.
- ✓ The objective of regression analysis is to find the relationship (or approximate relationship) between variables.
- ✓ A regression model is a mathematical equation that describes the relationship between two or more variables.
- ✓ Correlation analysis is is a statistical tool that is concerned with the question of whether there is a relationship between variables.
- ✓ In other words, correlation analysis determines how strong the linear relationship is between the X and Y variables.

1. SIMPLE LINEAR REGRESSION

- ✓ The relationship between two variables in a regression analysis is expressed by a mathematical equation called a regression equation or model.
- ✓ A regression equation, when plotted, may assume one of many possible shapes, including a straight line.
- ✓ However, a regression equation that gives a straight-line relationship between two variables is called a simple **linear regression model**, otherwise, the model is called a nonlinear regression model.
- ✓ Thus, to estimate the relationship between a response or dependent variable (Y) and an explanatory or independent variable (X), we use the simple linear regression model given by $y_i = \alpha + \beta x_i + \varepsilon_i\,,\ i = 1, 2, 3, \dots, n$

where

$$x = the\ value\ of\ the\ independent\ variable$$

$$y = the\ value\ of\ the\ dependent\ variable$$

$$\alpha = y\ axis\ intercept\ parameter$$

$$\beta = slope\ parameter\ of\ a\ straight\ line$$

$$\varepsilon_i = random\ error\ \ and$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

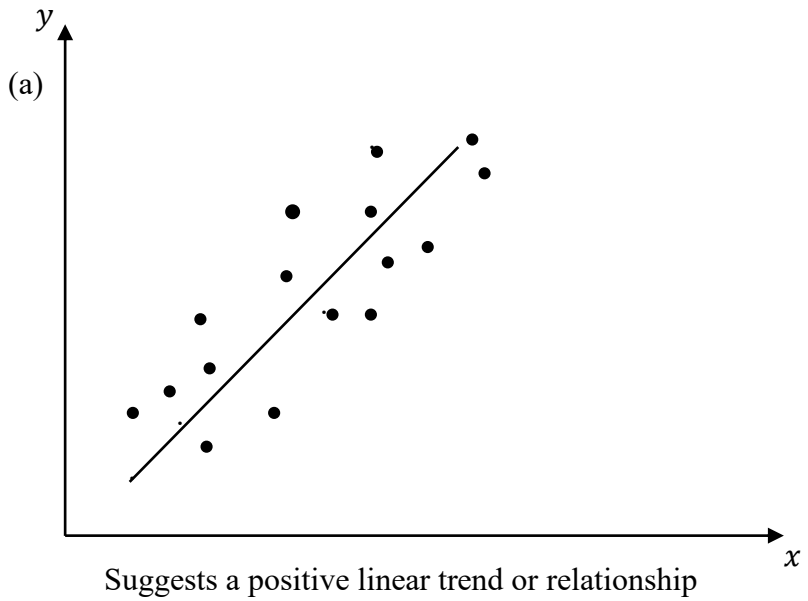
- ✓ The regression model stated above is called a **probabilistic model** or a **statistical relationship.**

NOTE

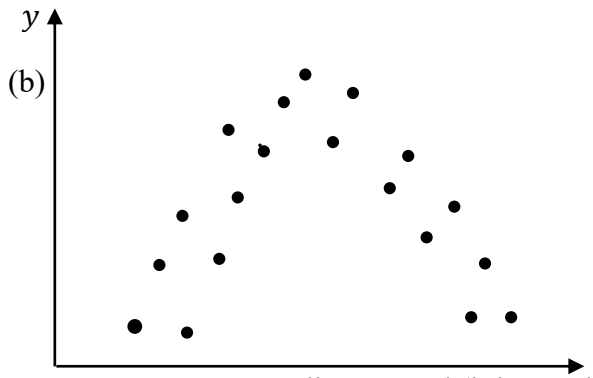
1. A regression model is a mathematical equation that describes the relationship between two or more variables.
2. A simple regression model includes only two variables, that is, one independent and one dependent variables.
3. The dependent variable is the one being explained, and the independent variable is the one that explains
4. The simple linear regression model is only appropriate if a scatter plot of X versus Y suggests a linear trend (or linear relationship).

e.g. the plots

(a)



Suggests a positive linear trend or relationship



Does not suggest a linear trend (it is non-linear)

- ✓ In general, Linear regression is conducted using the following steps

  Step 1. Identify the predictor or independent variable denoted by X and the dependent or response variable which needs to be estimated usually denoted by Y. It can be useful to construct a table containing the known variable values.

  Step 2. Construct a scatter plot

  A scatter plot is a graph of the variable values and can be useful in visually determining/detecting the relationship between the two variables.

  Step 3. Calculate the linear regression equation or model given by

$$y_i = \alpha + \beta x_i + \varepsilon_i , \qquad i = 1, 2, 3, \dots, n$$

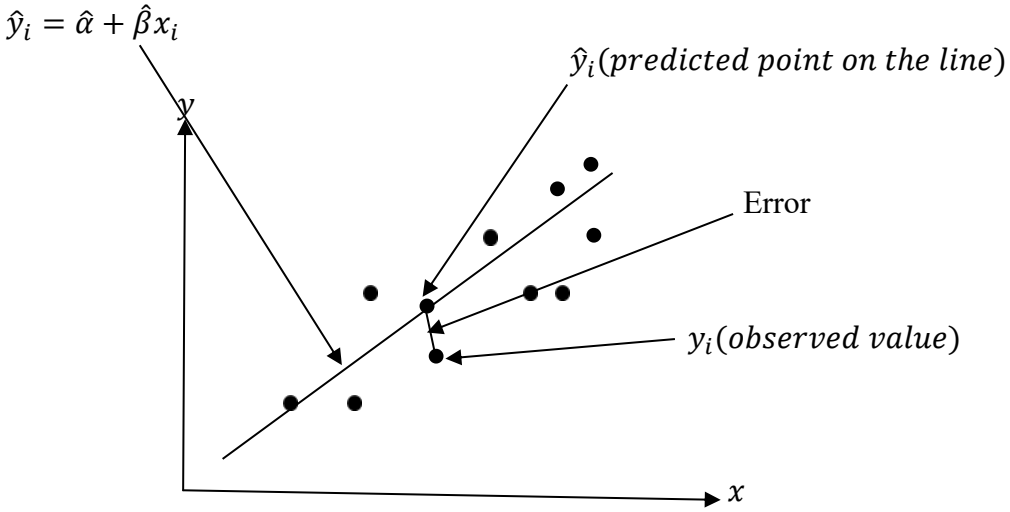
  Step 4. Make estimations from the regression equation

  Use the straight-line equation to estimate values of Y from values of X and establish the validity of these estimations.

2.1. LEAST SQUARES ESTIMATION

- ✓ Regression analysis uses the method of least squares to find the straight-line equation of best fit.
- ✓ In the regression model: $y_i = \alpha + \beta x_i + \varepsilon_i$, α and β are the **population parameters**.
- ✓ So, the regression line obtained for the above model by using the population data is called the **population regression line**.
- ✓ The values of α and β in the population regression line are called the **true values of the $y-$intercept and slope**, respectively.
- ✓ However, population data are difficult to obtain. As a result, we almost always use sample data to estimate the regression model.

- ✓ The values of the $y-$intercept and slope calculated from sample data on x and y are called the estimated values of α and β and are denoted by $\hat{\alpha}$ and $\hat{\beta}$, respectively.
- ✓ Using $\hat{\alpha}$ and $\hat{\beta}$, we write the estimated regression model as: $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$
- ✓ Now, let $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ be the equation of a fitted regression line, we need to find the equation of the line that best fits the data by minimizing the squares of the residuals or errors, i.e.



- ✓ $\varepsilon_i = y_i - \hat{y}_i$ are the errors i.e. the difference between the actual observations and what the model predicts. Thus,

$$\sum \varepsilon_i = \sum (y_i - \hat{y}_i) = 0$$

- ✓ Hence, to find the line that best fits the scatter of points, we cannot minimize the sum of errors, instead, we minimize the **error sum of squares**, denoted by **SSE**, which is obtained by adding the squares of errors. Thus,

$$L = SSE = \sum_{i=1}^{n} \varepsilon_i^2 = \sum (y_i - \hat{y}_i)^2$$

The least squares method gives the values of $\hat{\alpha}$ and $\hat{\beta}$ for the estimated regression model: $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ such that the sum of squared errors (SSE) is minimized.

- ✓ And so, we minimize the sum of squared errors by differentiating with respect to α and β as follows:

$$
\begin{aligned}
L &= \sum_{i=1}^{n} \varepsilon_i^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\
&= \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2 \\
\frac{\partial L}{\partial \alpha} &= 2 \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)(-1) \\
&= -2 \sum_{i=1}^{n} (y_i - \alpha - \beta x_i) \\
\frac{\partial L}{\partial \alpha} &= 0
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \sum_{i=1}^{n} (y_i - \alpha - \beta x_i) &= 0 \\
n\bar{y} - n\alpha - n\beta\bar{x} &= 0 \\
n\alpha + n\bar{x}\beta &= n\bar{y} \qquad (1)
\end{aligned}
$$

$$
\begin{aligned}
\frac{\partial L}{\partial \beta} &= 2 \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)(-x_i) \\
&= -2 \sum_{i=1}^{n} (x_i y_i - \alpha x_i - \beta x_i^2) \\
\frac{\partial L}{\partial \beta} &= 0
\end{aligned}
$$

$$
\begin{aligned}
\Rightarrow \sum_{i=1}^{n} (x_i y_i - \alpha x_i - \beta x_i^2) &= 0 \\
\sum_{i=1}^{n} x_i y_i - n\alpha\bar{x} - \beta \sum_{i=1}^{n} x_i^2 &= 0 \\
n\bar{x}\alpha + \beta \sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i \qquad (2)
\end{aligned}
$$

Solving (1) and (2) simultaneously we have:

$$
\begin{aligned}
n\bar{x}\alpha + \beta\sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i \\
n\alpha + n\bar{x}\beta &= n\bar{y} \quad \rightarrow \times\ \bar{x}
\end{aligned}
$$

$$
\begin{aligned}
n\bar{x}\alpha + \beta\sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i \\
n\bar{x}\alpha + n\bar{x}^2\beta &= n\bar{x}\bar{y}
\end{aligned}
$$

$$
\begin{aligned}
\beta\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\beta &= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} \\
\beta\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right) &= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} \\
\hat{\beta} &= \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} \\
&= \frac{S_{xy}}{S_{xx}}
\end{aligned}
$$

from equation (1) we have

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Here

$$
\begin{aligned}
S_{xx} &= \sum_{i=1}^{n} (x_i - \bar{x})^2 \\
&= \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}
\end{aligned}
$$

$$
\begin{aligned}
S_{xy} &= \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \\
&= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} \\
&= \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n}
\end{aligned}
$$

EXAMPLE

A company official obtained the following information on annual salary in thousands of dollars (y) and years of experience (x) for 12 technicians at various plants of the company.

Years of experience (x)	12	16	6	23	27	8	5	19	23	13	16	8
Salary in thousands of kwacha (y)	29	31	23	34	38	24	22	34	36	27	33	27

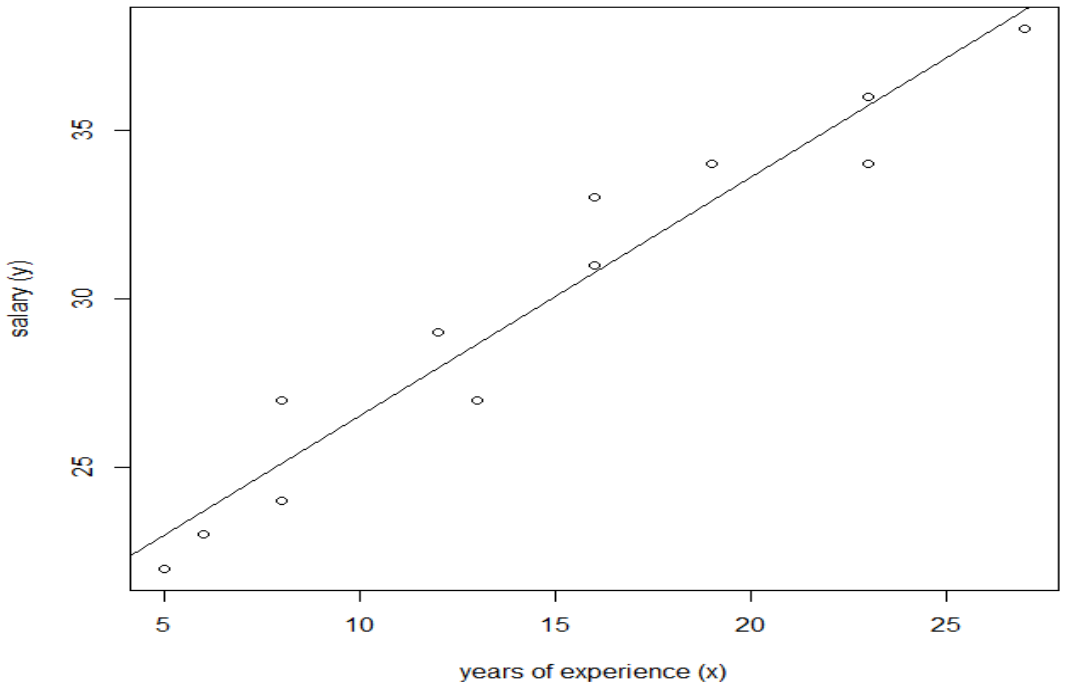
(a) Draw a scatter diagram. Is there a linear relationship between years of experience and annual salary?

(b) Estimate the linear regression line. Explain the meaning of the estimated parameters in the model.

(c) Estimate the expected annual salary of a technician with 15 years of experience.

Solution

(a)



Yes the scatter diagram suggests a linear relationship between years of experience and annual salary.

(b)

$$\sum_{i=1}^{n} x_i = 176 \ , \qquad \sum_{i=1}^{n} x_i^2 = 3162 \ , \qquad n = 12$$

$$\sum_{i=1}^{n} y_i = 358 \ , \qquad \sum_{i=1}^{n} y_i^2 = 10990 \ , \qquad \sum_{i=1}^{n} x_i y_i = 5661$$

Note that the above summary statistics can easily be obtained from your calculator.

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n} \\ &= 5661 - \frac{(176)(358)}{12} \\ &= 410.3333 \end{aligned}$$

$$\begin{aligned} S_{xx} &= \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n} \\ &= 3162 - \frac{(176)^2}{12} \\ &= 580.6667 \end{aligned}$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{410.3333}{580.6667} = 0.7067 = 0.71$$

$$\begin{aligned} \hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x} \\ &= \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta}\frac{(\sum_{i=1}^{n} x_i)}{n} \\ &= \frac{358}{12} - 0.7067\left(\frac{176}{12}\right) \\ &= 19.4684 \\ &= 19.47 \end{aligned}$$

$$\therefore \ \hat{y} = 19.47 + 0.71x$$

INTREPRETATION:

A technician with no experience receives an average income of 19.47 thousand Kwacha (K19,470).

The annual salary increases on average by 0.71 thousand Kwacha (K710) for each additional year of experience.

(c)

$$\begin{aligned}\hat{y} &= 19.47 + 0.71(15)\\ &= 30.12\ thousand\ Kwacha\\ &or\ K30{,}120\end{aligned}$$

2.2. INFERENCE ABOUT THE SLOPE PARAMETER

- ✓ Inference concerning the slope parameter can either be done using the F−test or the t −test as follows:

2.2.1. F – test

- ✓ To test the hypothesis

$H_0: \beta = 0 \quad against$
$H_1: \beta \neq 0$

we partition the total variation in y as follows:

$Total\ variation = explained\ variation + unexplained\ variation$

Or $SST = SSR + SSE$

where

$$\begin{aligned}SST &= Total\ sum\ of\ squares\\ &= \sum_{i=1}^{n}(y_i - \bar{y})^2\\ &= \sum_{i=1}^{n} y_i^2 - \frac{(\sum_{i=1}^{n} y_i)^2}{n}\\ &= S_{yy}\end{aligned}$$

$$\begin{aligned}SSR &= Regression\ sum\ of\ squares\\ &= \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2\\ &= \hat{\beta}S_{xy}\end{aligned}$$

$$\begin{aligned}SSE &= Error\ sum\ of\ squares\\ &= \sum_{i=1}^{n}(y_i - \hat{y}_i)^2\\ &= SST - SSR\\ &= S_{yy} - \hat{\beta}S_{xy}\end{aligned}$$

Prepare the ANOVA table:

Source of variation (Source)	Sum of squares (SS)	Degrees of freedom (df)	Mean squares (MS)	F^*
Regression	SSR	1	$MSR = SSR$	$\frac{MSR}{MSE}$
Error	SSE	$n-2$	$MSE = \frac{SSE}{n-2}$	
Total	SST	$n-1$		

We reject H_0 if $F^* > f_\alpha(1, n-2)$

NOTE

1. $MSR = SSR$ since $MSR = \frac{SSR}{1}$.
2. MSE is used to estimate σ^2.
3. We can easily verify that

$$\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 = \hat{\beta}S_{xy}$$

i.e.

$$\begin{aligned}\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2 &= \sum_{i=1}^{n}\left(\hat{\alpha} + \hat{\beta}x_i - \bar{y}\right)^2 = \sum_{i=1}^{n}\left(\bar{y} - \hat{\beta}\bar{x} + \hat{\beta}x_i - \bar{y}\right)^2 \\ &= \sum_{i=1}^{n}\left(-\hat{\beta}\bar{x} + \hat{\beta}x_i\right)^2 = \sum_{i=1}^{n}\hat{\beta}^2(x_i - \bar{x})^2 \\ &= \hat{\beta}^2\sum_{i=1}^{n}(x_i - \bar{x})^2 = \hat{\beta}^2 S_{xx} = \hat{\beta}\frac{S_{xy}}{S_{xx}}S_{xx} = \hat{\beta}S_{xy} \\ &= \hat{\beta}S_{xy}\end{aligned}$$

2.2.2. T – test

- ✓ To test the hypothesis

$$\begin{aligned}&H_0: \beta = \beta_0 \quad against \\ &H_a: \beta < \beta_0 \quad or \\ &H_1: \beta \neq \beta_0 \quad or \\ &H_A: \beta > \beta_0\end{aligned}$$

we use the test statistic

$$T = \frac{\hat{\beta} - \beta_0}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}}$$

which is a t – distribution with $n - 2$ degrees of freedom, where $\hat{\sigma}^2 = MSE$.

- ✓ We reject H_0 against

$$H_a: \beta < \beta_0 \quad if \quad t^* < -t_{\alpha,n-2}$$
$$H_1: \beta \neq \beta_0 \quad if \quad |t^*| > t_{\frac{\alpha}{2},n-2}$$
$$H_A: \beta > \beta_0 \quad if \quad t^* > t_{\alpha,n-2}$$

- ✓ Note that the T – test is more general as specific values of β_0 can be tested while the F – test is only used to test $\beta = 0$.

2.2.3. CONFIDENCE INTERVAL

- ✓ A $100(1-\alpha)\%$ confidence interval for β in the regression line $y = \alpha + \beta x$ is given by

$$\hat{\beta} \pm t_{\frac{\alpha}{2},n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

2.2.4. COEFFICIENT OF DETERMINATION

- ✓ The ratio

$$R^2 = \frac{explained\ variation}{total\ variation} = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

called the coefficient of determination gives the proportion of variation in y values explained by the linear relationship with x (the regression line).

- ✓ In general, the coefficient of determination measures the extent to which the independent variable determines the dependent variable.
- ✓ The closer the points are to the line, the greater the value of R^2.

3. CORRELATION ANALYSIS

- ✓ A regression is only useful if the X and Y values show some degree of linear relationship (i.e. a cluster of points about a nonhorizontal line).

- ✓ The degree of linear relationship between the X and Y values is measured by the linear correlation coefficient also known as Karl Pearson's correlation coefficient or simply Pearson's correlation.
- ✓ The Pearson's correlation coefficient typically denoted by r is given by

$$r = \frac{n(\sum xy) - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2] \times [n\sum y^2 - (\sum y)^2]}}$$

where:

r is the sample correlation coefficient

x is the values of the independent variable

y is the values of the dependent variable

n is number of observations

- ✓ The Pearson's correlation coefficient is some time called the bivariate or pairewise because its primary duty is to determine the relationship between two variables or detecting a relationship between a pair of variables, i.e., (X, Y).
- ✓ The correlation analysis can also be computed by using the concept of the methods of least squares, that is,

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

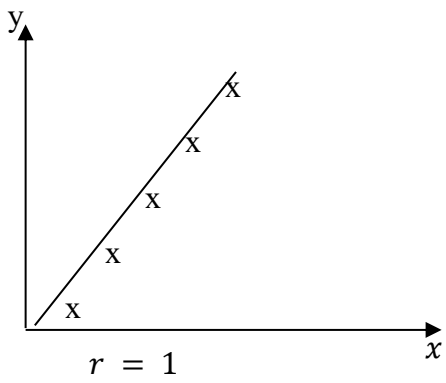
where

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) \\ &= \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y} \\ &= \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n} \\ S_{xx} &= \sum_{i=1}^{n}(x_i - \bar{x})^2 \\ &= \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n} \end{aligned}$$

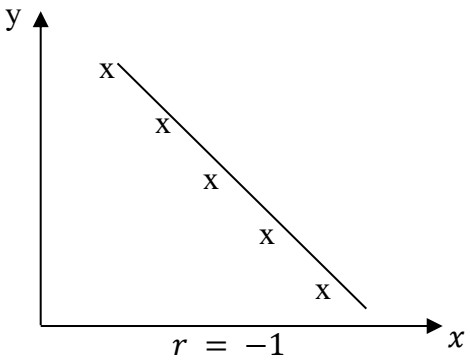
✓ The above stated correlation coefficient is also called the ***Pearson product moment correlation coefficient.***

3.1. PROPERTIES OF r

1. The value of r is always between -1 and 1 i.e. $-1 \leq r \leq 1$.
2. If $r = 1$, then it is said to be a perfect positive linear correlation and in such a case, all points in the scatter diagram lie exactly on a straight line with a positive slope.
3. If the values are closer to the absolute value of 1, it is an indication that there is a strong relationship between the variables being correlated.
4. Similarly if $r = -1$, then the correlation is said to be a perfect negative linear correlation and, in this case, all points in the scatter diagram lie on a straight line with a negative slope.
5. If the values are closer to the absolute value of -1, it is an indication that there is a weak relationship between the variables being correlated.

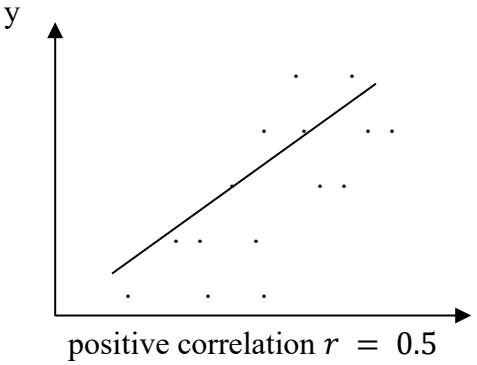


perfect positive correlation

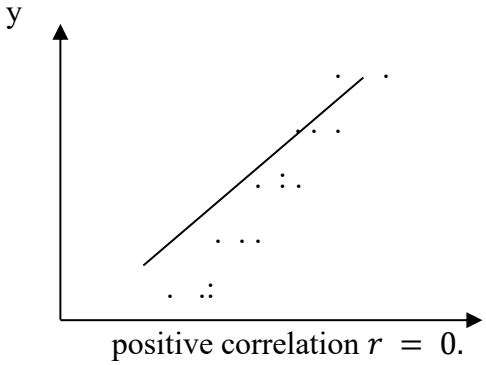


perfect negative correlation

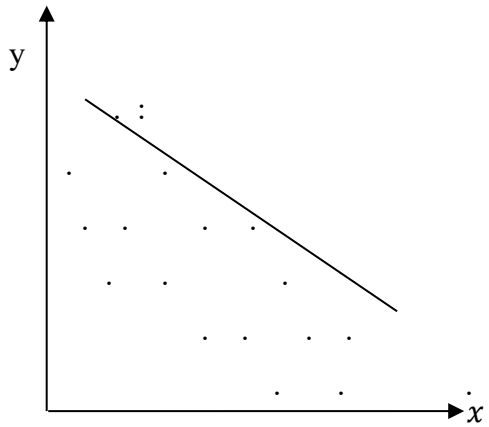
6. Suppose $r \neq 0$, then for a positive r, the line has a positive slope and for a is negative r, the line has a negative slope.
7. The closer r is to 1 or -1, the closer the points tend to cluster about the line.



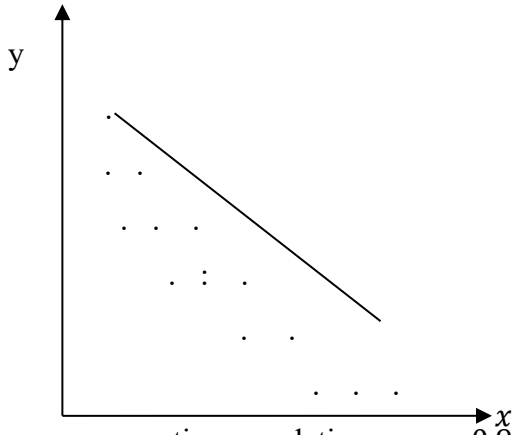
positive correlation $r = 0.5$



positive correlation $r = 0.$

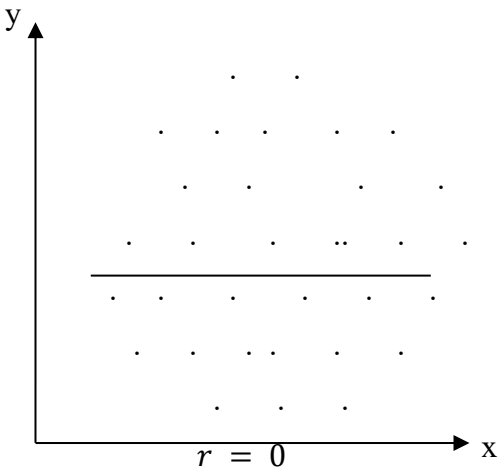


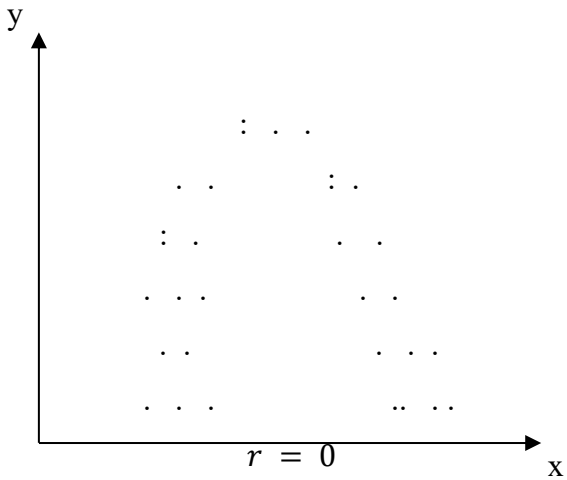
negative correlation $r = -0.6$



negative correlation $r = -0.9$

8. If $r = 0$, then there is no linear correlation between the two variables x and y. And that the points are scattered all over the diagram, as shown in the diagrams below.
9. If the values are closer to zero, it is an indication that there is little or no relationship between the variables being correlated.



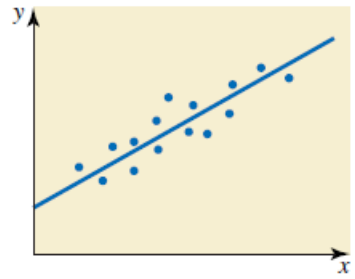


NOTE

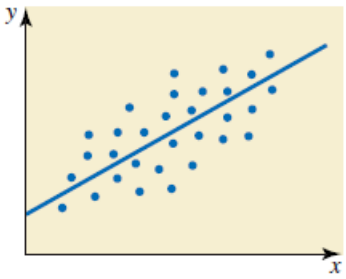
1. The sign of a correlation coefficient describes the type of the relationship between variables being correlated, for instance:
   (a) A positive correlation indicates that there is a positive linear relationship between the variables: as one variable increases in value so does the other, e.g., if the number of classes attended by a student increase in value, so do the test grades/scores.

(b) A negative correlation indicates that there is a negative linear relationship between the variables: as one variable increases in value the other variable decreases in value, e.g, when a student increased a number of missed class, that would result in test scores typically decrease.

2. If the correlation between two variables is positive and close to 1, we say that the variables have **a strong positive linear correlation** as shown in diagram (a).
3. If the correlation between two variables is positive but close to zero, then the variables have **a weak positive linear correlation** as shown in diagram (b).

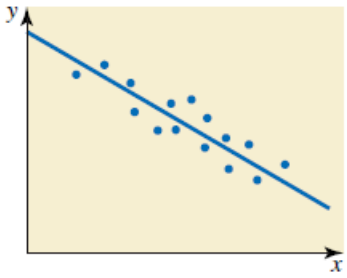


(*a*) Strong positive linear correlation
(*r* is close to 1)

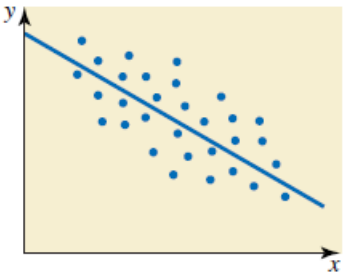


(*b*) Weak positive linear correlation
(*r* is positive but close to zero)

4. In contrast, if the correlation between two variables is negative and close to −1, then the variables are said to have **a strong negative linear correlation** as shown in diagram (c).
5. If the correlation between two variables is negative but close to zero, there exists **a weak negative linear correlation** between the variables as shown in diagram (d).



(*c*) Strong negative linear correlation
(*r* is close to –1)



(*d*) Weak negative linear correlation
(*r* is negative and close to zero)

6. Graphically, a strong correlation indicates that the points in the scatter diagram are very close to the regression line, and a weak correlation indicates that the points in the scatter diagram are widely spread around the regression line.

3.1.1. SUMMARY FOR THE RANGE OF CORRELATION COEFFICIENT

(1) 0.00 to 0.19	Very weak positive
(2) 2. 0.20 to 0.39	Weak positive
(3) 0.40 to 0.59	Moderate positive
(4) 0.60 to 0.79	Strong positive
(5) 0.80 to 1.00	Very strong positive
(6) 0.00 to -0.19	very weak negative
(7) -0.20 to -0.39	Weak negative
(8) -0.40 to -0.59	Moderate negative
(9) -0.60 to -0.79	Strong negative
(10) -0.80 to -1.00	Very strong negative

3.2. ASSUMPTIONS

(1) Both variables should be continuous variables,

(2) Both variables should be independent

3.3. OTHER TYPES OF CORRELATION COEFFICIENTS

- ✓ Other types of correlation coefficients include:

1. Partial correlation coefficients. It is a correlation that should be used when you are measuring the association between two variables but want to factor out the effect of one or more other variables.
    - *For example, if there are three variables 'A', 'B', 'Z', if you want to find the relationship between 'A' and 'B' with the influence of 'Z' being controlled you can use partial correlation*.
2. Rank correlation coefficients also known as the spearman's correlation analysis denoted by rho is simply the Pearson's correlation of the ranking of the raw data.
3. Kendall correlation measures how well the relationship between two variables can be described using a monotonic function, that is, the variables move in the same direction (increasing or decreasing) but not necessarily at a constant rate.

3.4. COEFFICIENT OF DETERMINATION

- ✓ The square of the correlation coefficient (r^2) gives the coefficient of determination, which was explained above.
- ✓ That is, $r^2 = R^2 \Longrightarrow r = \sqrt{R^2}$ where $R^2 = coefficient\ of\ determination$.

EXAMPLES

1. A farmer was interested in the relationship between the amount of fertilizer (x) and the number of bushels (y) of soybeans produced. The farmer conducted an experiment and obtained the following data:

Fertilizer in hundreds of kgs per acre (x)	1.0	2.5	3.0	3.0	3.4	4.0	4.0	4.5
Bushels per acre (y)	25	32	35	32	35	39	41	40

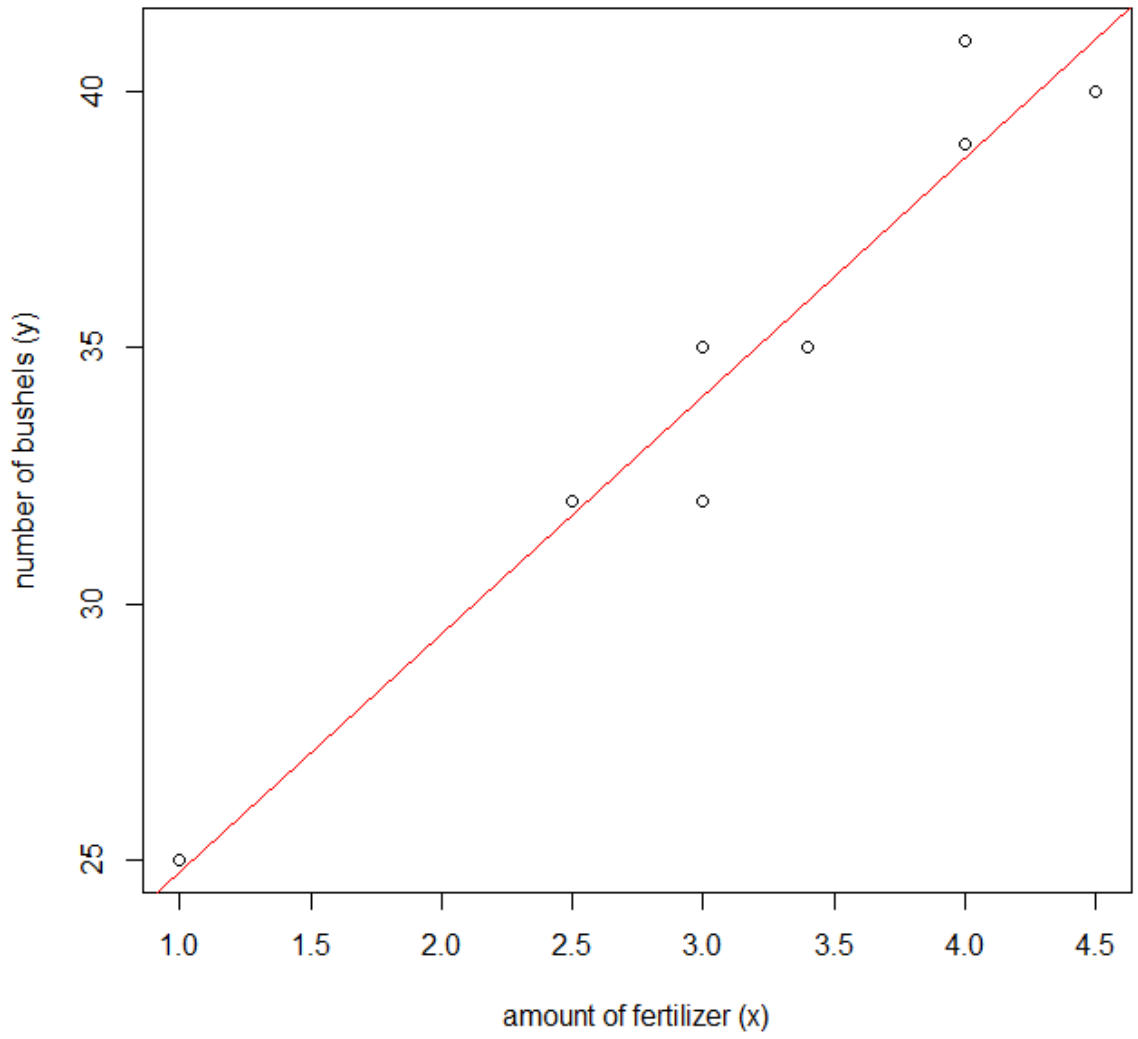
   (a) Draw a scatter diagram. Do the data appear to be positively or negatively correlated?
   (b) Find the equation of the regression line. Explain the meaning of the regression line.
   (c) Test the hypothesis that $\beta = 0$ against the alternative that $\beta \neq 0$. (i.e. Is there a significant linear relationship between x and y?)
   (d) Construct a 95% confidence interval for β.
   (e) Find the coefficient of determination and interpret the value obtained.
   (f) Find the correlation coefficient.

$$\left[\sum_{i=1}^{n} x_i = 25.4\,, \quad \sum_{i=1}^{n} x_i^2 = 89.06\,, \quad \sum_{i=1}^{n} y_i = 279\,, \quad \sum_{i=1}^{n} y_i^2 = 9925\,, \quad \sum_{i=1}^{n} x_i y_i = 925\,, \quad n = 8\right]$$

SOLUTION

(a)

From the plot the variables are positively correlated. The number of bushels increases with the amount of fertilizer per acre.



(b)

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n} \\ &= 925 - \frac{(25.4)(279)}{8} \\ &= 39.175 \end{aligned}$$

$$S_{xx} = \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n}$$
$$= 89.06 - \frac{(25.4)^2}{8}$$
$$= 8.415$$
$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{39.175}{8.415} = 4.655377$$
$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$
$$= \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta}\frac{(\sum_{i=1}^{n} x_i)}{n}$$
$$= \frac{279}{8} - 4.655377\left(\frac{25.4}{8}\right)$$
$$= 20.094$$
$$\therefore \ \hat{y} = 20.094 + 4.655x$$

The average expected number of bushels when no fertilizer is applied is 20.094. The average expected number of bushels increases by 4.655 for every unit increase (100kg) in the number of fertilizer per acre (i.e. the number of bushels increases by 4.655 for every additional 100kg of fertlizer applied per acre).

(c) Using an F – test

Testing problem

$H_0: \beta = 0$
$H_1: \beta \neq 0$

Calculation

$$SST = S_{yy}$$
$$= \sum_{i=1}^{n} y_i^2 - \frac{(\sum_{i=1}^{n} y_i)^2}{n}$$
$$= 9925 - \frac{279^2}{8}$$
$$= 194.875$$

$$SSR = \hat{\beta}S_{xy}$$
$$= 4.655377(39.175)$$
$$= 182.374$$

$$SSE = SST - SSR$$
$$= 194.875 - 182.374$$
$$= 12.501$$

Source	SS	df	MS	F^*
Regression	182.374	1	182.374	87.53
Error	12.501	6	2.0835	
Total	194.875	7		

$$F^* = 87.53 > f_{0.05}(1,6) = 5.99$$

Conclusion

Reject H_0 at the 5% level of significance, therefore β is significantly different from 0 i.e. there is a significant linear relationship between amount of fertilizer applied (x) and the number of bushels obtained (y).

OR (Using a T – test)

Testing problem

$H_0: \beta = 0$
$H_1: \beta \neq 0$

Critical region

Reject H_0 if $|t^*| > t_{\frac{\alpha}{2},n-2} = t_{0.025,6} = 2.447$

Calculation

$$T = \frac{\hat{\beta} - \beta_0}{se(\hat{\beta})} = \frac{\hat{\beta}}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} \quad (since\ \beta_0 = 0)$$

$$t^* = \frac{4.655}{\sqrt{\frac{2.0835}{8.415}}}$$
$$= 9.355$$
$$i.e. |t^*| > 2.447$$

Conclusion

Reject H_0 at the 5% level of significance, therefore β is significantly different from 0 i.e. there is a significant linear relationship between amount of fertilizer applied (x) and the number of bushels obtained (y).

NOTE

$$\begin{aligned} t^2_{\frac{\alpha}{2},n-2} &= F_\alpha(1, n-2) \\ i.e.\ t^2_{0.025,6} &= f_{0.05}(1,6) \\ (t^*)^2 &= F^* \end{aligned}$$

(d)

$$\hat{\beta} \pm t_{\frac{\alpha}{2},n-2} se(\hat{\beta})$$

$$4.655 \pm t_{0.025,6}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$4.655 \pm 2.447\sqrt{\frac{2.0835}{8.415}}$$

$$(3.437, 5.873) \ \ or \ \ 3.437 < \beta < 5.873$$

(e)

$$R^2 = \frac{SSR}{SST} = \frac{182.374}{194.875} = .936$$

This means that 93.6% of the variation in the number of bushels is explained by the amount of fertilizer applied (i.e. by the linear relationship). About 6% of unexplained variability may be due to chance or other variables not considered in the model.

(f)

$$\begin{aligned} r &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \\ &= \frac{39.175}{\sqrt{8.415(194.875)}} \\ &= 0.9674 \end{aligned}$$

OR

$$\begin{aligned} r &= \sqrt{R^2} \\ &= \sqrt{\frac{SSR}{SST}} \\ &= \sqrt{\frac{182.374}{194.875}} \\ &= 0.9674 \end{aligned}$$

NOTE

Note that to use $r = \sqrt{R^2}$ we are actually finding $r = sign\sqrt{R^2}$. Since we know there is a positive linear correlation between the two variables (from the plot in (a)), the sign will be positive. If the plot showed a negative correlation the sign of the correlation coefficient would be negative.

2. Consider the following data:

x	4	4	6	6	8	8
y	12	11	10	9	7	6

(a) Draw a scatter diagram. Does the model $y = \alpha + \beta x$ seem appropriate?

(b) Estimate the linear regression line. Explain the meaning of the parameter(s) in the model.

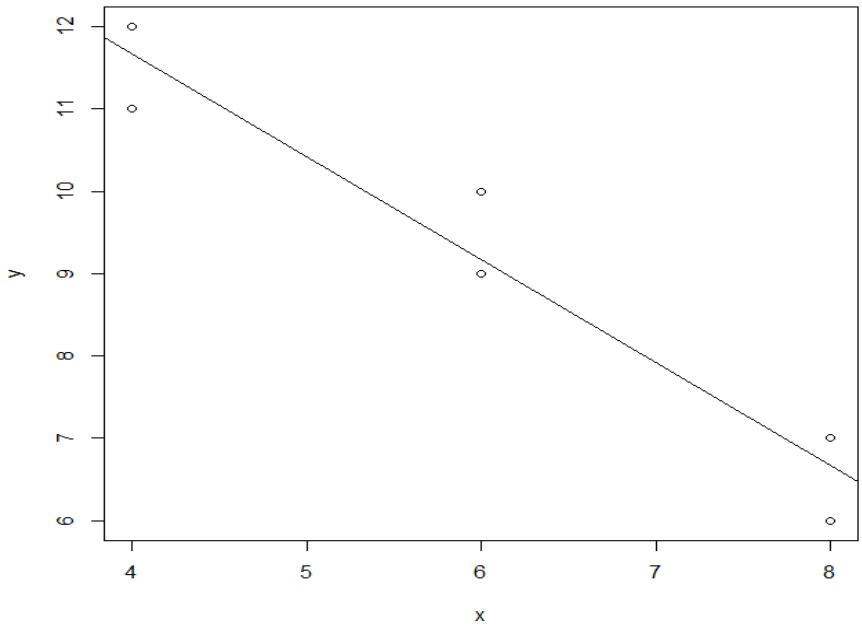
(c) Would we conclude that there is a negative linear relationship between between the x and y values of the population? Use $\alpha = 0.01$.

(d) Find a 99% confidence interval for the slope parameter β.

(e) Find the correlation coefficient.

SOLUTION

(a)



Yes the model seems appropriate since there's a linear trend between x and y.

(b)

$$\sum_{i=1}^{n} x_i = 36 \qquad \sum_{i=1}^{n} x_i^2 = 232 \qquad n = 6$$

$$\sum_{i=1}^{n} y_i = 55 \qquad \sum_{i=1}^{n} y_i^2 = 531 \qquad \sum_{i=1}^{n} x_i y_i = 310$$

$$\hat{y} = \hat{\alpha} + \hat{\beta}x$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}}$$

$$\begin{aligned} S_{xy} &= \sum_{i=1}^{n} x_i y_i - \frac{(\sum_{i=1}^{n} x_i)(\sum_{i=1}^{n} y_i)}{n} \\ &= 310 - \frac{36(55)}{6} \\ &= -20 \end{aligned}$$

$$\begin{aligned} S_{xx} &= \sum_{i=1}^{n} x_i^2 - \frac{(\sum_{i=1}^{n} x_i)^2}{n} \\ &= 232 - \frac{(36)^2}{6} \\ &= 16 \end{aligned}$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \frac{-20}{16} = -1.25$$

$$\begin{aligned} \hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x} \\ &= \frac{\sum_{i=1}^{n} y_i}{n} - \hat{\beta}\frac{(\sum_{i=1}^{n} x_i)}{n} \\ &= \frac{55}{6} - (-1.25)\left(\frac{36}{6}\right) \\ &= 16.6667 \end{aligned}$$

$$\therefore \ \hat{y} = 16.67 - 1.25x$$

On average y reduces by 1.25 when x increases by one unit.

Note: the intercept has no specific meaning in this case.

(c)

1. Testing problem

$H_0: \beta = 0$
$H_1: \beta < 0$

2. Test statistic

$$T = \frac{\hat{\beta} - \beta_0}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}}$$

3. Critical region

Reject H_0 if

$$t^* < -t_{\alpha,n-2} = -t_{0.01,4} = -3.747$$

4. Calculation

$$\hat{\sigma}^2 = MSE = \frac{SSE}{n-2}$$

$$SSE = S_{yy} - \hat{\beta} S_{xy}$$

$$\begin{aligned} S_{yy} &= \sum_{i=1}^{n} y_i^2 - \frac{(\sum_{i=1}^{n} y_i)^2}{n} \\ &= 531 - \frac{55^2}{6} \\ &= 26.833333 \end{aligned}$$

$$SSE = 26.8333333 - (-1.25)(-20) = 1.8333333$$

$$\hat{\sigma}^2 = \frac{1.833333}{4} = 0.4583333$$

$$t^* = \frac{\hat{\beta} - \beta_0}{\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}} = \frac{-1.25 - 0}{\sqrt{\frac{0.4583333}{16}}} = -7.385$$

i.e. $t^* < -3.747$

5. Conclusion

Reject H_0 at $\alpha = 0.01$ level of significance. Therefore we conclude that there is a significant linear relationship between x and y.

(d)

$$\hat{\beta} \pm t_{\frac{\alpha}{2},n-2}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

$$-1.25 \pm t_{0.005,4}\sqrt{\frac{0.45833333}{16}}$$

$$-1.25 \pm 4.604\sqrt{\frac{0.45833333}{16}}$$

$$(-2.029, -0.471)$$

(e)

$$\begin{aligned} r &= \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \\ &= \frac{-20}{\sqrt{16(26.833333)}} \\ &= -0.965 \end{aligned}$$

OR

$$\begin{aligned} r &= sign\sqrt{R^2} \\ &= -\sqrt{1-\frac{SSE}{SST}} \\ &= -\sqrt{1-\frac{SSE}{S_{yy}}} \\ &= -\sqrt{1-\frac{1.8333333}{26.833333}} \\ &= -0.965 \end{aligned}$$

NOTE

The sign of the correlation in this example is negative since there is a negative linear relationship between x and y.